EXPERIMENT-1

```
> f <- c(200,450,300,1500,700,44)
> L <- c(1,5,15,20,50,80)
> U <- c(5,15,20,50,80,110)
> cf <- cumsum(f)
> N <- sum(f)
> i <- which(cf >= N/2)[1]
> m <- L[i] + ((N/2 - cf[i-1]) / f[i]) * (U[i]-L[i])
> print(m)
[1] 32.94
>
```

EXPERIMENT-2

```
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,4$
> mean(x)
[1] 29.96296
> median(x)
[1] 25
> names(which.max(table(x)))
[1] "25"
> (min(x)+max(x))/2
[1] 41.5
> quantile(x,c(0.25,0.75))
 25%  75%
20.5 35.0
>
```

EXPERIMENT-3

```
20.5 35.0
> x <- c(200,300,400,600,1000)
> (x-min(x))/(max(x)-min(x))
[1] 0.000 0.125 0.250 0.500 1.000
> (x-mean(x))/sd(x)
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
> 
```

EXPERIMENT-4

```
> x <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75)
> b <- split(x, rep(1:3, each=8))
> lapply(b, function(x) rep(mean(x),length(x)))
$`1`
[1] 15.25 15.25 15.25 15.25 15.25 15.25 15.25 15.25

$`2`
[1] 22.625 22.625 22.625 22.625 22.625 22.625 22.625 22.625

$`3`
[1] 58.25 58.25 58.25 58.25 58.25 58.25 58.25 58.25

> lapply(b, function(x) rep(median(x),length(x)))
$`1`
[1] 15 15 15 15 15 15 15 15

$`2`
[1] 21.5 21.5 21.5 21.5 21.5 21.5 21.5 21.5

$`3`
[1] 58 58 58 58 58 58 58 58

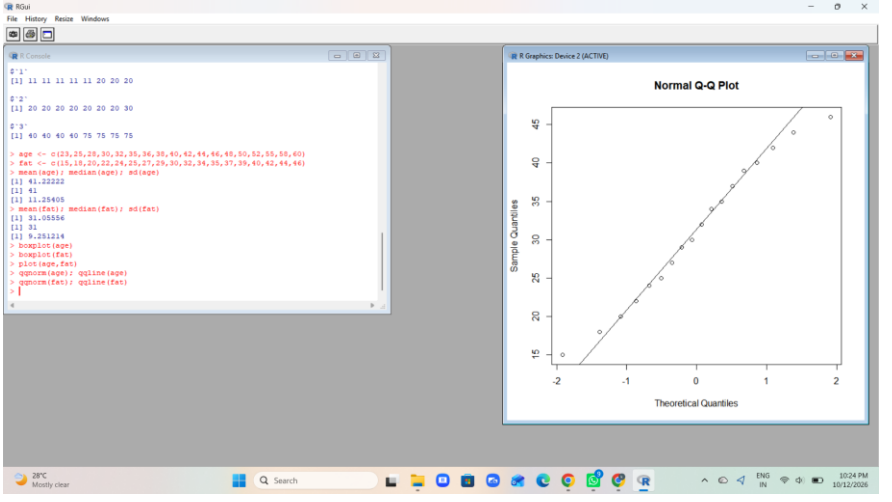
> lapply(b, function(x) ifelse(abs(x-min(x))<=abs(x-max(x)),min(x),max(x)))
$`1`
[1] 11 11 11 11 11 20 20 20

$`2`
[1] 20 20 20 20 20 20 20 30

$`3`
[1] 40 40 40 40 75 75 75 75

> 
```

EXPERIMENT-5



EXPERIMENT-6

```
> boxplot(age)
> boxplot(fat)
> plot(age,fat)
> qqnorm(age); qqline(age)
> qqnorm(fat); qqline(fat)
> age <- c(23,25,28,30,32,35,36,38,40,42,44,46,48,50,52,55,58,60)
> (35-min(age))/(max(age)-min(age))
[1] 0.3243243
> (35-mean(age))/12.94
[1] -0.4808518
> 35/100
[1] 0.35
>
```

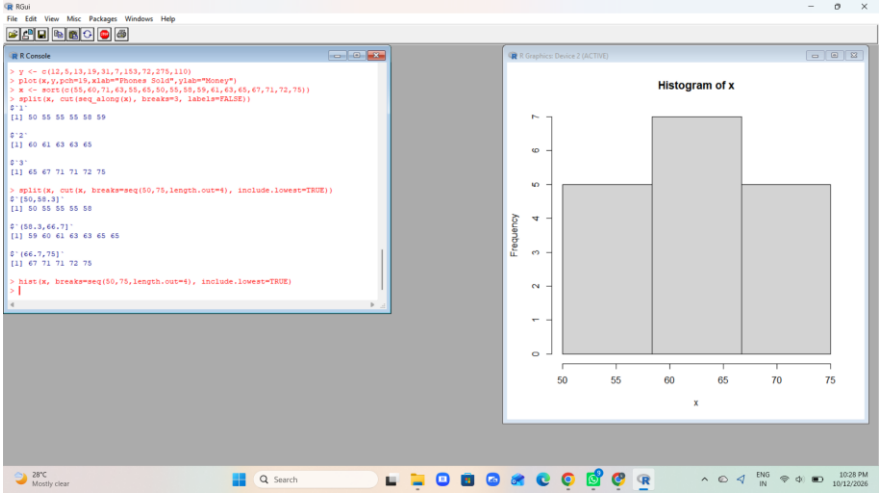
EXPERIMENT-7

```
[1] 0.3243243
> (35-mean(age))/12.94
[1] -0.4808518
> 35/100
[1] 0.35
> x <- c(9,25,23,12,11,6,7,8,9,10)
> mean(x)
[1] 12
> median(x)
[1] 9.5
> names(which.max(table(x)))
[1] "9"
>
```

EXPERIMENT-8

```
[1] 0.35
> x <- c(9,25,23,12,11,6,7,8,9,10)
> mean(x)
[1] 12
> median(x)
[1] 9.5
> names(which.max(table(x)))
[1] "9"
> x <- c(4,1,5,7,10,2,50,25,90,36)
> y <- c(12,5,13,19,31,7,153,72,275,110)
> plot(x,y,pch=19,xlab="Phones Sold",ylab="Money")
>
```

EXPERIMENT-9



EXPERIMENT-10

```
> hist(x, breaks=seq(50,75,length.out=4), include.lowest=TRUE)
> x <- c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
> IQR(x)
[1] 4.975
> sd(x)
[1] 4.445835
>
```

EXPERIMENT-11

```
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,4$
> quantile(x,c(0.25,0.75))
 25%  75%
20.5 35.0
>
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,4$
> quantile(x,c(0.25,0.75))
 25%  75%
20.5 35.0
>
```